

# DSBDL Viva Question Bank

Data Science and Big Data Laboratory — Complete Answers

## 1) What is Pandas library in Python?

Pandas is an open-source data manipulation and analysis library for Python. It provides fast, flexible, and expressive data structures like Series (1D) and DataFrame (2D) designed to make working with structured (tabular, multidimensional, potentially heterogeneous) data intuitive and easy. It is built on top of NumPy and is widely used in data science and machine learning workflows.

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## 2) List some key features of Pandas.

Key features of Pandas include:

- DataFrame and Series data structures for easy data handling
  - Handling of missing data (NaN values)
  - Flexible data reshaping and pivoting
  - Merging and joining datasets
  - Time series functionality
  - Data filtering, grouping, and aggregation
  - Reading and writing data from/to CSV, Excel, SQL, JSON
  - Label-based and integer-based indexing
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## 3) What is NumPy Library in Python?

NumPy (Numerical Python) is a fundamental open-source library for scientific computing in Python. It provides support for large, multi-dimensional arrays and matrices, along with a collection of high-level mathematical functions to operate on these arrays efficiently. NumPy is the foundation for many other data science libraries like Pandas, SciPy, and Scikit-learn.

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## 4) What is Matplotlib library?

Matplotlib is a comprehensive Python library for creating static, animated, and interactive visualizations. It provides a MATLAB-like interface and allows creation of plots such as line charts, bar charts, histograms, scatter plots, pie charts, etc. The pyplot module is most commonly used. It integrates well with NumPy and Pandas.

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## 5) What is the difference between Seaborn and Matplotlib?

Matplotlib is a low-level, highly customizable plotting library that gives full control over every aspect of the visualization, but requires more code. Seaborn is built on top of Matplotlib and provides a higher-level interface with better default aesthetics and built-in themes. Seaborn is better for statistical visualizations (heatmaps, violin plots, pair plots), while Matplotlib is more general-purpose and flexible. Seaborn requires less code for attractive plots.

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## 6) Is Sklearn and Scikit-learn the same library? What is its use in data science?

Yes, sklearn and Scikit-learn are the same library. 'sklearn' is the Python import name while 'Scikit-learn' is the official name. It is the most widely used machine learning library in Python, providing simple and efficient tools for data mining and data analysis. It includes algorithms for classification, regression, clustering, dimensionality reduction, model selection, and preprocessing.

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## 7) What are common functions in Pandas and NumPy?

Pandas functions: `read_csv()`, `head()`, `tail()`, `info()`, `describe()`, `shape`, `isnull()`, `dropna()`, `fillna()`, `groupby()`, `merge()`, `sort_values()`, `value_counts()`, `drop_duplicates()`.

NumPy functions: `array()`, `zeros()`, `ones()`, `linspace()`, `arange()`, `reshape()`, `mean()`, `median()`, `std()`, `var()`, `dot()`, `sum()`, `min()`, `max()`, `sqrt()`, `random.rand()`.

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## 8) What is a DataFrame in Python?

A DataFrame is a two-dimensional, size-mutable, and heterogeneous tabular data structure in Pandas. It is similar to a spreadsheet or SQL table, with labeled rows (index) and columns. Each column can hold different data types (int, float, string, etc.). DataFrames are the most commonly used Pandas object.

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## 9) How to find duplicates in Python? (Python Command)

To find duplicates in a Pandas DataFrame:

- `df.duplicated()` — returns a boolean Series marking duplicate rows
  - `df[df.duplicated()]` — shows all duplicate rows
  - `df.duplicated(subset=['column_name'])` — finds duplicates in a specific column
  - `df.drop_duplicates()` — removes all duplicate rows
  - `df['col'].duplicated()` — finds duplicates in a single column
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## 10) What is the use of describe() command?

The `describe()` function in Pandas generates descriptive statistics of a DataFrame or Series. For numerical columns it returns: count, mean, standard deviation (std), minimum value, 25th percentile (Q1), 50th percentile (median/Q2), 75th percentile (Q3), and maximum value. For object/string columns it shows count, unique, top (most frequent), and frequency. Usage: `df.describe()`

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## 11) Which are the Naive Bayes classification algorithms used in Python?

Scikit-learn provides the following Naive Bayes classifiers:

- `GaussianNB` — for continuous/normally distributed features
- `MultinomialNB` — for discrete count features (e.g., text classification)
- `BernoulliNB` — for binary/boolean features

- ComplementNB — improved version of MultinomialNB for imbalanced datasets
  - CategoricalNB — for categorical features
- All are available under `sklearn.naive_bayes` module.
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## 12) What is the significance of Confusion Matrix?

A confusion matrix is a performance evaluation tool for classification models. It is a table that shows the number of correct and incorrect predictions broken down by each class. It helps to understand how well a model performs, where it makes mistakes, and which classes are being confused with each other. It forms the basis for metrics like accuracy, precision, recall, and F1-score.

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## 13) What is TP, TN, FP, FN in Confusion Matrix?

- TP (True Positive): Model predicted Positive and the actual value is Positive — correct prediction.
  - TN (True Negative): Model predicted Negative and the actual value is Negative — correct prediction.
  - FP (False Positive): Model predicted Positive but actual value is Negative — also called Type I error.
  - FN (False Negative): Model predicted Negative but actual value is Positive — also called Type II error.
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## 14) What is Recall?

Recall (also called Sensitivity or True Positive Rate) measures how many actual positive cases were correctly identified by the model. Formula:  $\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$ . A high recall means the model identifies most of the actual positives. It is important in cases where missing a positive is costly, such as disease detection.

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## 15) What is Precision?

Precision measures how many of the predicted positive cases are actually positive. Formula:  $\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$ . High precision means fewer false alarms (false positives). It is important when the cost of a false positive is high, such as spam detection or fraud detection.

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## 16) What is F1 Score?

F1 Score is the harmonic mean of Precision and Recall. Formula:  $\text{F1} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$ . It provides a single metric that balances both precision and recall, making it useful when there is an uneven class distribution. F1 score ranges from 0 (worst) to 1 (best).

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## 17) What is the need of Data Visualization in Data Science?

Data visualization is essential because: it helps understand patterns, trends, and outliers in data; it makes complex data more accessible and understandable; it supports decision-making

by presenting insights clearly; it helps detect relationships between variables; it is used to communicate findings to non-technical stakeholders; and it helps in exploratory data analysis (EDA) before building models.

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### 18) What is an Outlier?

An outlier is a data point that significantly differs from other observations in the dataset. It lies far outside the expected range of values. Outliers can occur due to measurement errors, data entry mistakes, or genuine rare events. They can negatively impact statistical analyses and machine learning models. Box plots and Z-score methods are commonly used to detect outliers.

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### 19) When to use Histogram and Pie Chart?

**Histogram:** Used to show the frequency distribution of a single continuous numerical variable. It displays the shape, spread, and center of data. Use when you want to understand how data is distributed (e.g., age distribution of customers).

**Pie Chart:** Used to show proportions or percentages of parts to a whole. Best used when comparing a few categories (ideally less than 5). Use when you want to show the relative share of each category (e.g., market share of companies).

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### 20) What are the challenges in Big Data Visualization?

- **Volume:** Visualizing millions or billions of data points is computationally expensive.
  - **Variety:** Data comes in different formats (text, images, audio) making unified visualization difficult.
  - **Velocity:** Real-time streaming data needs real-time rendering.
  - **Complexity:** Multi-dimensional data is hard to represent in 2D/3D visuals.
  - **Noise:** Large datasets contain noise that can obscure meaningful patterns.
  - **Scalability:** Tools must scale to handle big data efficiently.
  - **Interpretation:** Misleading visuals can lead to wrong decisions.
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### 21) What is Joint Plot and Dist Plot?

**Joint Plot:** A Seaborn function (`sns.jointplot()`) that displays the relationship between two variables using a scatter plot along with the distribution of each variable shown in the margins as histograms or KDE plots.

**Dist Plot:** A Seaborn function (`sns.distplot()` or `sns.histplot()`) that shows the distribution of a single variable using a histogram combined with a Kernel Density Estimate (KDE) curve.

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### 22) What are the tools used for Data Visualization?

- **Python libraries:** Matplotlib, Seaborn, Plotly, Bokeh, Altair
- **R libraries:** ggplot2
- **Business Intelligence tools:** Tableau, Power BI, Qlik Sense
- **Web-based:** D3.js, Google Data Studio

- Others: Apache Superset, Grafana, Kibana (for big data dashboards)
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### 23) What is Data Wrangling?

Data Wrangling (also called Data Munging) is the process of cleaning, transforming, and restructuring raw data into a usable format for analysis. It includes handling missing values, removing duplicates, fixing incorrect data types, merging datasets, reshaping data, and filtering irrelevant records. It is a critical step in the data science workflow that often takes 60-80% of a data scientist's time.

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### 24) What is Data Transformation?

Data transformation refers to converting data from one format, structure, or scale to another to make it suitable for analysis or model training. It includes: normalization (scaling data to [0,1]), standardization (zero mean and unit variance), encoding categorical variables (label encoding, one-hot encoding), log transformation (to handle skewed data), and aggregation.

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### 25) What is the use of StandardScaler function in Python?

StandardScaler from sklearn.preprocessing is used to standardize features by removing the mean and scaling to unit variance (z-score normalization). Formula:  $z = (x - \text{mean}) / \text{std}$ . This ensures that all features have a mean of 0 and a standard deviation of 1, which is important for algorithms sensitive to feature scales like SVM, KNN, and logistic regression.

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### 26) What is Hadoop?

Hadoop is an open-source framework developed by Apache for distributed storage and processing of large datasets (Big Data) across clusters of computers using simple programming models. It is based on the MapReduce programming model and the Hadoop Distributed File System (HDFS). It can scale from a single server to thousands of machines and is fault-tolerant.

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### 27) What is HDFS and MapReduce?

HDFS (Hadoop Distributed File System): The primary storage system of Hadoop that stores large files across multiple machines in a cluster. It splits files into blocks (default 128 MB), replicates them (default 3 copies) for fault tolerance, and provides high-throughput access to data.

MapReduce: A programming model and processing engine in Hadoop for parallel processing of large datasets. It works in two phases — Map (splits data and processes in parallel) and Reduce (aggregates/summarizes results from all mappers).

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### 28) What are the components of the Hadoop Ecosystem?

- HDFS — Distributed file system for storage
- MapReduce — Data processing framework
- YARN — Resource manager and job scheduling

- Hive — SQL-like querying on Hadoop data
  - HBase — NoSQL database on HDFS
  - Pig — Scripting platform for data flow
  - Sqoop — Data transfer between Hadoop and relational databases
  - Flume — Log data ingestion
  - ZooKeeper — Coordination service
  - Spark — Fast in-memory data processing
  - Oozie — Workflow scheduler
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## **29) What is Scala?**

Scala (Scalable Language) is a modern, general-purpose programming language that combines object-oriented and functional programming paradigms. It runs on the Java Virtual Machine (JVM) and is fully interoperable with Java. It is widely used in big data processing, especially with Apache Spark, which is written in Scala.

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## **30) What are the features of Scala?**

- Combines object-oriented and functional programming
  - Statically typed with powerful type inference
  - Runs on JVM — interoperable with Java
  - Immutability support with `val` keyword
  - Pattern matching for concise code
  - Higher-order functions and lambda expressions
  - Built-in concurrency support with Akka
  - Traits for mixin-based composition
  - Concise and expressive syntax
  - Strong support for Big Data (Apache Spark)
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## **31) How is Scala different from Java?**

- Scala supports functional programming; Java is primarily object-oriented.
  - Scala has type inference; Java requires explicit type declarations in most cases.
  - Scala is more concise — fewer lines of code needed.
  - Scala supports immutability by default (`val`); Java uses `final` keyword.
  - Scala has pattern matching; Java does not (until recent versions).
  - Scala does not have static members; it uses object (singleton) instead.
  - Scala has a more expressive type system.
  - Scala has no checked exceptions; Java enforces them.
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## **32) List applications of Scala.**

- Big Data processing with Apache Spark
- Web development using Play Framework and Lift
- Distributed systems with Akka

- Kafka (messaging platform built in Scala)
  - Backend services and microservices
  - Data engineering and ETL pipelines
  - Financial systems requiring high concurrency
  - Scientific computing
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### 33) What is Data Science?

Data Science is an interdisciplinary field that uses scientific methods, algorithms, processes, and systems to extract knowledge and insights from structured and unstructured data. It combines statistics, mathematics, computer science, and domain expertise to analyze data and build predictive models. Applications include recommendation systems, fraud detection, healthcare analytics, and business intelligence.

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### 34) What is Big Data?

Big Data refers to extremely large and complex datasets that cannot be processed or analyzed using traditional data processing tools or databases. It involves capturing, storing, managing, and analyzing massive volumes of data from various sources such as social media, sensors, transactions, and logs. It is typically processed using distributed frameworks like Hadoop and Spark.

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### 35) What are the characteristics of Big Data?

The 5 Vs of Big Data:

- Volume — Huge amounts of data (terabytes to petabytes)
  - Velocity — Speed at which data is generated and processed
  - Variety — Different types of data (structured, semi-structured, unstructured)
  - Veracity — Uncertainty and quality of the data
  - Value — Usefulness and insight derived from the data
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### 36) List the phases in Data Science Life Cycle.

1. Problem Definition — Understand the business problem
  2. Data Collection — Gather relevant data from various sources
  3. Data Cleaning / Wrangling — Handle missing, inconsistent, or noisy data
  4. Exploratory Data Analysis (EDA) — Analyze and visualize data
  5. Feature Engineering — Create and select relevant features
  6. Model Building — Train machine learning models
  7. Model Evaluation — Assess model performance using metrics
  8. Model Deployment — Deploy the model to production
  9. Monitoring — Track model performance over time
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### 37) What is Central Tendency?

Central tendency is a statistical measure that identifies a single value as representative or central to a dataset. It describes the 'center' or 'typical value' in a distribution. The three main measures of central tendency are: Mean (arithmetic average), Median (middle value), and Mode (most frequent value).

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### 38) What is Dispersion?

Dispersion (or Spread) measures how spread out or scattered the data values are around the central tendency. It tells us about the variability in the dataset. Common measures include: Range, Variance, Standard Deviation, Interquartile Range (IQR), and Mean Absolute Deviation (MAD). Low dispersion means data points are close together; high dispersion means they are spread far apart.

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### 39) What is Mean, Mode, Mid-Range, Median? Calculate for: 10, 22, 13, 10, 21, 43, 77, 21, 10

Dataset: 10, 22, 13, 10, 21, 43, 77, 21, 10

Sorted: 10, 10, 10, 13, 21, 21, 22, 43, 77

Mean = Sum / n = (10+22+13+10+21+43+77+21+10) / 9 = 227 / 9 ≈ 25.22

Median = Middle value of sorted data = 5th value = 21

Mode = Most frequent value = 10 (appears 3 times)

Mid-Range = (Max + Min) / 2 = (77 + 10) / 2 = 87 / 2 = 43.5

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### 40) What is Variance?

Variance measures how far each number in the dataset is from the Mean, and thus from every other number. Formula: Variance ( $\sigma^2$ ) =  $\sum (x_i - \mu)^2 / N$  (population) or  $\sum (x_i - \bar{x})^2 / (n-1)$  (sample). A higher variance indicates that data points are spread widely around the mean. Variance is the square of standard deviation.

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### 41) What is Standard Deviation?

Standard Deviation ( $\sigma$  or  $s$ ) is the square root of variance. It measures the average distance of each data point from the mean. Formula:  $\sigma = \sqrt{\text{Variance}}$ . It has the same unit as the original data, making it more interpretable than variance. A small standard deviation means data is clustered around the mean; a large value indicates greater spread.

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### 42) What is the Posterior Probability in Naive Bayes?

Posterior Probability is the probability of a hypothesis (class) being true given the observed evidence (features). In Naive Bayes:  $P(\text{Class} | \text{Features}) = P(\text{Features} | \text{Class}) \times P(\text{Class}) / P(\text{Features})$ . It is calculated after observing the data (updated belief). Naive Bayes classifies a new sample to the class with the highest posterior probability.



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#### 43) What is the Likelihood Probability in Naive Bayes?

Likelihood Probability is  $P(\text{Features} \mid \text{Class})$  — the probability of observing the given features given that the data belongs to a specific class. It measures how probable the observed data is under a particular hypothesis/class. In Naive Bayes, features are assumed to be conditionally independent, so the likelihood is calculated as the product of individual feature probabilities.

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#### 44) How to deal with Missing Values?

Methods to handle missing values:

- Remove rows/columns: `df.dropna()` — suitable when missing data is small
  - Mean/Median/Mode imputation: `df.fillna(df.mean())`
  - Forward fill / Backward fill: `df.fillna(method='ffill')`
  - Custom value fill: `df.fillna(0)` or `df.fillna('Unknown')`
  - Interpolation: `df.interpolate()`
  - Predictive imputation: Use ML models to predict missing values
  - KNN Imputer from Scikit-learn for sophisticated imputation
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#### 45) What is NLTK?

NLTK (Natural Language Toolkit) is a leading Python library for working with human language data (text). It provides tools for tokenization, stemming, lemmatization, parsing, tagging, classification, and semantic reasoning. It includes over 50 corpora and lexical resources including WordNet. It is widely used for Natural Language Processing (NLP) research and development.

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#### 46) What is Tokenization in NLP?

Tokenization is the process of breaking a text (sentence, paragraph, or document) into smaller units called tokens, which can be words, sentences, or subwords. For example: 'I love data science' → ['I', 'love', 'data', 'science']. NLTK provides `word_tokenize()` and `sent_tokenize()` for word-level and sentence-level tokenization respectively.

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#### 47) What is Stemming?

Stemming is the process of reducing a word to its root or base form by removing suffixes, without considering the context. For example: 'running' → 'run', 'studies' → 'studi'. NLTK provides PorterStemmer, LancasterStemmer, and SnowballStemmer. The result may not always be a valid English word, but it groups similar words together for analysis.

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#### 48) What is Lemmatization?

Lemmatization is similar to stemming but more sophisticated — it reduces a word to its base form (lemma) considering the context and part of speech. It always produces a valid dictionary word. For example: 'running' → 'run', 'better' → 'good'. NLTK uses WordNetLemmatizer. Lemmatization is slower but more accurate than stemming.

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#### **49) What is Corpus in NLP?**

A Corpus (plural: Corpora) is a large and structured collection of text or speech used for linguistic analysis, training NLP models, and building language resources. Examples include the Brown Corpus, Reuters Corpus, WordNet, and Wikipedia dumps. NLTK provides access to many standard corpora. A corpus serves as the training data for NLP tasks like text classification and sentiment analysis.

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#### **50) What is Spark Framework?**

Apache Spark is an open-source, distributed computing framework designed for fast, large-scale data processing. Unlike Hadoop MapReduce, Spark performs in-memory computation, making it up to 100x faster. It supports multiple languages (Python/PySpark, Scala, Java, R). Spark includes libraries for SQL (Spark SQL), streaming (Spark Streaming), machine learning (MLlib), and graph processing (GraphX).

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